



Lecture 3: Dispatch and variable generation

Energy Economics of the Green Transition

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Today

A workshop on Hirth (2013): what happens to the value of wind as wind is built, and whether that is anyone's problem.

		min
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Breaks of fifteen minutes after part 1 and after part 2. The worksheet is on paper; the notebook is optional.

Intro

Energy System Models, sections 2 and 3; appendix A if the duality machinery is new.

Hirth (2013), *The market value of variable renewables*. Sections 1, 3, 4.3, 5.1–5.5 and 7 are the ones today's questions point at.

Optional: Joskow (2011) on why levelised cost is the wrong comparison; López Prol et al. (2020) for cannibalisation in market data.

The question

Hirth (2013): what is a MWh of wind worth, compared with the average MWh, and how does that change as wind is built?

His answer, three ways:

source	wind's value factor	solar's
literature review	1.1 near zero share, falling	falls faster
regression, 2001–2010 market data	−1.62 per point of share (thermal)	
European model, fixed fleet	0.5 at a 30% share	< 0.5 at 15%
European model, long run	0.65 at 30%	

And a claim: profile, balancing and grid-related costs “are not market failures”.

Today: **why** (part 1), **how much**, and **what drives it** (part 2), **so what** (part 3).

Setup: two hours, three plants

A calm hour and a windy hour of equal length, nothing linking them. Wind capacity K is the knob.

	capacity, GW	marginal cost, EUR/MWh	availability (calm, windy)
Wind	K	0	0, 1
Coal	2	40	1, 1
Gas	2	80	1, 1
Load, GW	calm 1.5, windy 3		

Predict before you compute: at a small K , is wind's value factor above or below one? Write your guess down, with the reason.

The value of wind in a two-hour system

1. The dispatch problem, its conditions, the price in each hour for every K .
2. $\bar{p}_w(K)$, $\bar{p}(K)$, $VF_w(K)$; sketch; name the kinks.
3. Hirth's time-weighted base price against the note's.
4. Short, mid or long term? $\partial C^*/\partial K$, and the capture price as a cost saving.
5. General T hours: monotonicity, and the sign of $VF_w - 1$ at zero share.

Stop point

$VF_w(K)$ with its kinks named, and the derivative of question 4.

The full questions and the stretch are on the worksheet.

Solutions 1

Wind capacity rises from 2 to 2.5 GW. The price in the windy hour

1. falls, because more zero-cost supply entered
2. is unchanged, because coal is still the interior plant
3. falls to zero, because wind is now marginal

Solutions 1, Q1–Q2: the questions

- Q1. Set up the two-hour dispatch and derive its conditions. The price in each hour for every $K \geq 0$; the values of K where the price-setting plant changes.
- Q2. $\bar{p}_w(K)$, $\bar{p}(K)$ and $VF_w(K)$ for every K . Sketch $VF_w(K)$ and name the event behind each kink.

Solutions 1, Q1–Q2: the price, and the value factor

Windy hour, residual load $3 - K$; the calm hour is coal at 40 throughout.

	interior plant	λ_2	$\bar{p}_w$	$\bar{p}$	VF_w
$K < 1$	gas	80	80	66.7	1.2
$1 < K < 3$	coal	40	40	40	1.0
$K > 3$	wind	0	0	13.3	0

Two kinks: at $K = 1$ the last MW of gas leaves the windy hour; at $K = 3$ the last MW of coal leaves, and curtailment starts.

Between the kinks nothing happens: from $K = 1.5$ to 2.5 only coal's output changes.

- Q3. Hirth's base price is the time-weighted average; the note's is consumption-weighted. Both value factors at $K = 0.5$, and the general condition under which Hirth's is the larger.

Solutions 1, Q3: Hirth's base price against the note's

At $K = 0.5$: the time average is $\langle \lambda \rangle = (40 + 80)/2 = 60$, so Hirth's value factor is $80/60 = 1.33$ against the note's 1.2.

In general, with $\bar{D}$ the average load,

$$\bar{p} - \langle \lambda \rangle = \frac{1}{T\bar{D}} \sum_h (D_h - \bar{D}) \lambda_h = \frac{\text{Cov}(D, \lambda)}{\bar{D}}.$$

Hirth's is the larger whenever load and price move together, which is the normal case: load is what pushes the price up the merit order.

- Q4. Which of Hirth's short, mid and long term is this exercise? For T hours with a fixed fleet: $\partial C^*/\partial K = -\sum_h \gamma_h \lambda_h$ where wind is not curtailed, and the capture price as the cost saving per MWh of wind.
- Q5. Show $\bar{p}_w$ is non-increasing in K . The exact condition on the profiles for $VF_w > 1$ as $K \rightarrow 0$; what Hirth's correlation effect correlates with what.

Solutions 1, Q4–Q5: the envelope, and the board

Short term: the fleet is fixed and K is a parameter. Envelope theorem on wind's capacity constraint:

$$\frac{\partial C^*}{\partial K} = - \sum_h \gamma_h \bar{\mu}_{w,h} = - \sum_h \gamma_h \lambda_h \quad (\text{uncurtailed: } \bar{\mu}_{w,h} = \lambda_h).$$

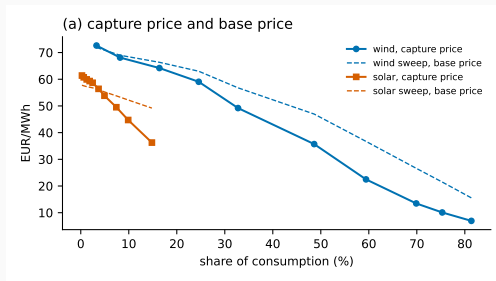
Per MWh delivered that is $\bar{p}_w$: the capture price is the marginal cost saving.

On the board:

- **Q4.** Short, mid or long term? $\partial C^* / \partial K$ by the envelope theorem; the capture price as the cost saving per MWh.
- **Q5.** $\bar{p}_w$ non-increasing in K ; the exact condition for $VF_w > 1$ as $K \rightarrow 0$, and what is correlated with what.

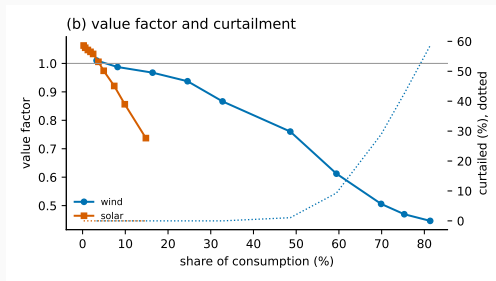
Part 2

Setup: the evidence (a)



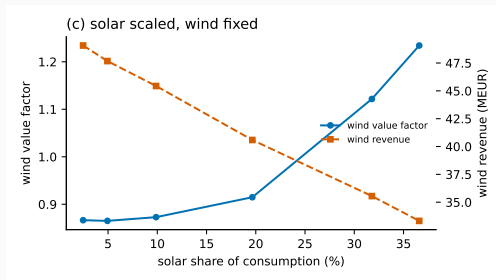
- The note's model on the ten-plant fleet, DK1 2024 profiles, every eighth hour; nothing stored or exported.
- Wind scaled by s (blue), solar by s (vermillion); one solve per point, placed at the technology's own share of consumption.
- Solid: the capture price $\bar{p}_g$. Dashed: the base price $\bar{p}$ along the same sweep. The gap is what the value factor measures.

Setup: the evidence (b)



- The same sweeps as a ratio, $VF_g = \bar{p}_g / \bar{p}$, with one marked.
- Dotted, right axis: curtailment, the share of the energy the weather offered that the system could not use.
- Where curtailment starts is on the figure; what the value factor does before and after it is question 2.

Setup: the evidence (c)



- The third experiment: solar scaled up, wind held at its base capacity, and *wind's* statistics recorded.
- Blue, left axis: wind's value factor. Vermillion, right axis: wind's revenue over the sampled year, million EUR.
- The worksheet's table has the numbers behind every point; the notebook reproduces them, optionally.

The cannibalisation curve, and what drives it

1. Wind's slopes, 3–33% and 33–59%, against Hirth's regression; two reasons, with directions.
2. Falling without curtailment, then with it: the price-setting plant in each regime.
3. Solar's slope against wind's; the profile statistic that explains it.
4. Solar up, wind fixed: $\Delta \ln VF_w = \Delta \ln \bar{p}_w - \Delta \ln \bar{p}$; absolute and relative.

Stop point

Two slopes for wind, one ratio for solar, the two terms of the decomposition.

The full questions and the stretch are on the worksheet.

Solutions 2

Solutions 2, Q1–Q3: the questions

- Q1. Wind's slope over 3–33% and 33–59%, against Hirth's regression. Two reasons his numbers and the model's differ, and which way each pushes.
- Q2. Falling without curtailment, then with it: the price-setting plant in each regime, and why the two are one mechanism. Hirth's Fig. 11 for this fleet at $s = 4$. What curtailment does to the capacity factor and to revenue per MW.
- Q3. Solar's slope against wind's, and the statistic in Table 2 that explains it.

Solutions 2, Q1–Q3: the slopes against Hirth's

Slope of the value factor, percentage points per percentage point of share:

wind, 3–33%	−0.47
wind, 33–59%	−1.0
solar, 0–15%	−2.2
Hirth, thermal systems	−1.62
Hirth, hydro systems	−0.26

- **Q1. Below Hirth's thermal estimate:** this merit order is flatter.
- **Q1. Pushing the other way:** over his decade the fleet adjusts around the wind, which flattens his slope; ours holds the fleet fixed.
- **Q2. Two regimes:** for $s > 1.5$ wind is curtailed: prices drop to zero, and the capacity factor falls with them.
- **Q3. Solar falls five times faster:** it is concentrated in the same midday hours.

- Q4. Solar up, wind fixed, $s = 1 \rightarrow 24$: decompose $\Delta \ln VF_w = \Delta \ln \bar{p}_w - \Delta \ln \bar{p}$. Which term is absolute and which relative cannibalisation? Which number does a wind investor read, and which a policy maker who indexes support to the value factor?

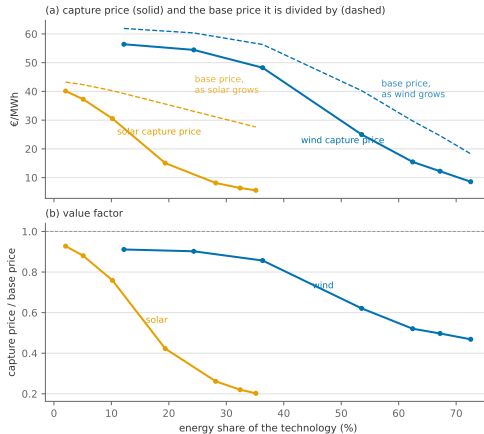
Solutions 2, Q4: the ratio trap

Solar up, wind fixed, $s = 1 \rightarrow 24$:

	$s = 1$	$s = 24$	$\Delta \ln$
$\bar{p}_w$	49.2	43.5	-0.12
$\bar{p}$	56.8	35.3	-0.48
VF_w	0.87	1.23	+0.35
wind revenue, MEUR	49	33	-0.39

- **Decomposition:** $\Delta \ln VF_w = -0.12 - (-0.48) = +0.35$. The ratio rose because its denominator fell faster.
- **Absolute cannibalisation** is the capture price, down 12%. **Relative** is the value factor, up 35%.
- **The investor** reads the capture price and the revenue: both fell. **A support scheme indexed to the value factor** reads a number that improved.

Closing part 2: the same experiment on the real DK1 fleet (note, figure 3.5)



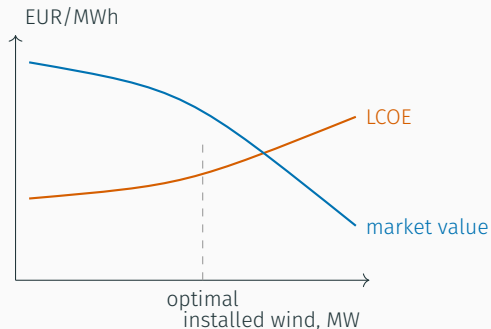
Closing part 2: what today's model leaves out, and where it returns

Every gap between this curve and Hirth's is a missing kind of flexibility, and each one is a later lecture.

left out today	wind's value factor	where it returns
exports: surplus is curtailed instead	lower	lecture 5
storage, flexible demand	lower	lecture 4
a fleet that adjusts as wind grows	lower than long run	lecture 6

Part 3

Setup: Hirth's figure 3, and his claim



Rising cost, falling value, and the intersection as the optimal amount.

“Profile, balancing, and grid-related costs are not market failures, but represent the intrinsic lower value of electricity during times of high supply, at remote sites, and the economic costs of uncertainty.”

Wind costs c^l per MW and year; $\Gamma = \sum_h \gamma_h$.

Optimal deployment, and whether the drop is a market failure

1. Hirth's figure from the model: free entry, the planner's condition, a subsidy σ , the loss.
2. The developer's argument: which of Hirth's three costs, market failure or not; \$3.5 against his sentence.
3. Joskow: what is the good, why is a MWh not a MWh; $K = 0.5$ and $K = 3.5$.

Stop point

The free-entry condition, and one paragraph placing the developer's argument on a named result.

The full questions and the stretch are on the worksheet.

Solutions 3

- Q1. Hirth's figure 3 from the model: the free-entry condition in terms of the capture price; show it is the planner's condition for K . A production subsidy σ : how far past the optimum, and which area is the loss.

Solutions 3, Q1: the intersection is the optimum

Free entry stops where one MW just covers its cost,

$$\Gamma \bar{p}_w(K) = c^l \iff \bar{p}_w(K) = LCOE_w.$$

The planner sets $-\partial C^*/\partial K = c^l$, which by part 1 is the same equation (Result 6).

A subsidy σ per MWh moves entry to $\bar{p}_w + \sigma = LCOE_w$; the loss is the area between the two curves right of the intersection.

Solutions 3, Q2–Q3: the questions

- Q2. The developer's argument: which of Hirth's three costs, and is it a market failure? Against his sentence, which two of his costs *are* market failures in \$3.5, under what condition, and what the targeting principle prescribes.
- Q3. Joskow in the model's terms: what is the good, and why a MWh is not a MWh. The two-hour system at $K = 0.5$ and $K = 3.5$.

Solutions 3, Q2–Q3: the drop is pecuniary

Q2, profile cost is the developer's argument: the price doing its job (Result 1). Pecuniary, not a market failure, and compensating it overshoots.

Q2, grid and balancing are real externalities *when* unpriced: congestion under flat or zonal tariffs, reserve socialised. Nodal pricing; imbalance charges. One instrument per externality.

Q3, Joskow: the good is electricity in hour h . Wind's MWh is worth 80 at $K = 0.5$ and 0 at $K = 3.5$; coal's 57 and 40.

Wrap-up

What carries forward

1. **The price is a staircase in residual load.** Wind walks it down in the hours it produces, and is paid in exactly those hours. Result 1, hour by hour; the capture price is the marginal cost saving.
2. **The slope of the merit order is the sufficient statistic.** Every driver Hirth tests works through it or through the price level.
3. **Cannibalisation is pecuniary.** What it measures is missing flexibility: nothing in today's model can move a MWh from a surplus hour to a tight one.

Lecture 4 adds the first thing that can: storage. The question becomes what a battery is worth, and to whom.

Before lecture 4 (Monday 28 September): *Energy System Models*, section 4; Zerrahn et al. (2018) and Ambec and Crampes (2021), the two papers the workshop works through. Full solutions to today's worksheet go up after class.

- Ambec, Stefan and Claude Crampes (2021). **“Real-Time Electricity Pricing to Balance Green Energy Intermittency”**. *Energy Economics* 94, p. 105074.
- Hirth, Lion (2013). **“The Market Value of Variable Renewables: The Effect of Solar Wind Power Variability on Their Relative Price”**. *Energy Economics* 38, pp. 218–236.
- Joskow, Paul L. (2011). **“Comparing the Costs of Intermittent and Dispatchable Electricity Generating Technologies”**. *American Economic Review: Papers & Proceedings* 101.3, pp. 238–241.
- López Prol, Javier, Karl W. Steininger, and David Zilberman (2020). **“The Cannibalization Effect of Wind and Solar in the California Wholesale Electricity Market”**. *Energy Economics* 85, p. 104552.
- Zerrahn, Alexander, Wolf-Peter Schill, and Claudia Kemfert (2018). **“On the Economics of Electrical Storage for Variable Renewable Energy Sources”**. *European Economic Review* 108, pp. 259–279.